

BUMPING UP AGAINST THE BARRIER WITH THE BINOMIAL METHOD

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The Cox-Ross-Rubinstein binomial model is a powerful and flexible tool for numerical valuation of different types of options. In the case of standard calls and puts, the convergence of the algorithm is swift. For short-maturity options and normal parameter values, accurate values are typically obtained within a hundred time steps.

In the case of barrier options, however, a naive application of the binomial lattice approach can lead to a seriously biased estimate even when a large number of time steps are

used. This note isolates the origin of the problem and indicates how to rectify it easily by using the right number of time steps in the tree. In one example, we show how a tree with twenty-one time steps can produce a much more accurate valuation than one with 800 time steps.

Problems with the naive binomial method have led to some inaccuracies in the literature. We provide corrected numbers and a comparison with accurate solution methods that have recently been discovered.

The variety of new option contracts has increased enormously in recent years. Many types of so-called exotic options are now popular items in the over-the-counter market. An important extension of the standard contract is to introduce a barrier that triggers some event that impacts the option's price. Down-and-out options are one example. In some cases analytical valuation equations can be found for these options, while in others numerical methods must be employed.

The standard Cox-Ross-Rubinstein binomial lattice is a useful valuation tool for standard options. Accurate values for short-dated options

are generally obtained within a hundred time steps.¹ This procedure can also be used to value options when there is a barrier, but the convergence of the method is very slow, and the results tend to have a persistent bias. This means that if a researcher uses the number of time steps that would give accurate values for standard options, the results for barrier options will be inaccurate and in general biased in a particular direction.

In this note we trace the source of this bias. It arises because the barrier will in general lie in between two adjacent layers of nodes in the lattice. The fact that the probability is discretized in this way

induces an error in the valuation of a barrier option. To correct it, we should arrange the number of time steps so that a horizontal layer of nodes is just beyond the barrier, and as close as possible to it.

We show how this can be done, and apply the method to value three different types of barrier options to illustrate its operation. These examples have been selected because there are analytic expressions for the limiting continuous-time cases to provide benchmarks for comparison purposes. The examples are also of interest because they represent important types of barrier options.

We show first that a naive application of the binomial method to down-and-out and down-and-in-call options leads to strongly biased results. Our approach, however, provides for good convergence. The naive binomial method has led to inaccurate results for capped options and vulnerable options in the literature. We provide corrected numbers. The latter are of current interest in connection with credit risk.

I. BARRIER OPTIONS

We illustrate the basic problem using barrier options as examples (European options, specifically). There are a wide variety of these options, and in many cases closed-form valuation solutions are available. Works by Rubinstein and Reiner [1993] and Rich [1993a] provide detailed formulas for many different types of barrier options.

A down-and-out call option is a simple example of a barrier option. This contract is like a standard call option with the proviso that if the asset value ever reaches a "barrier" price H below the current asset price during the option's lifetime the contract is cancelled. If the asset price always remains above the barrier, the payoff is the same as that of a standard call.

A down-and-in call becomes activated if the asset price ever reaches the barrier during the option's lifetime. Otherwise it is worthless at expiration, regardless of the final asset price.

A standard call option can be decomposed into a down-and-out call option and a down-and-in call option. Merton [1973] was the first to derive a closed-form solution for the down-and-out call option.

The notation is:

T = time to maturity of option;
 S = current market value of underlying asset;
 K = strike price of option;
 r = riskless rate continuously compounded;
 σ = standard deviation of rate of return on the asset;
 H = barrier level; and
 $BSC[S, K, \sigma, r, T]$ = Black-Scholes European call price.

We also make the standard Black-Scholes assumptions so that the asset price follows a geometric Brownian motion and markets are frictionless.

For $K > H$, the value of the down-and-in call option can be written as

$$DAIC[S, H, K, r, \sigma, T] =$$

$$\left(\frac{H}{S}\right)^{\gamma-1} BSC\left[\frac{H^2}{S}, K, r, \sigma, T\right]$$

where

$$\gamma = \frac{2r}{\sigma^2}$$

The value of the corresponding down-and-out call is obtained by subtracting this from the Black-Scholes value.

$$DAOC[S, H, K, r, \sigma, T] =$$

$$BSC[S, K, r, \sigma, T] - DAIC[S, H, K, r, \sigma, T]$$

Exhibit 1 provides numerical estimates of values of the standard European call option and both the down-and-in call option and the down-and-out call option for a range of values of n , where n is the number of time steps. Note that the convergence of the binomial model to its continuous-time Black-Scholes limit is quite rapid for the standard contract.

The situation is different for both the barrier options. The down-and-in call options converge very erratically, and their values always lie below the continuous-time limit. Even with fairly large values of n ,

EXHIBIT 1

NUMERICAL ESTIMATES OF STANDARD EUROPEAN CALL OPTION, DOWN-AND-IN CALL, AND DOWN-AND-OUT CALL BASED ON NAIVE BINOMIAL METHOD

Barrier Level = 90% of Strike Price

Parameters

S	=	95
K	=	100
σ	=	0.25 per year
T	=	1 year
r	=	0.10 per year
n	=	number of time divisions in binomial method

n	Standard Call Option	Down-and-In Call Option	Down-and-Out Call Option
25	11.5559	2.7073	8.8486
50	11.6729	4.4324	7.2405
75	11.6776	5.3775	6.3001
100	11.6342	4.1296	7.5045
150	11.6618	5.1007	6.5612
200	11.6658	4.4351	7.2307
400	11.6526	5.0022	6.6505
800	11.6559	5.0519	6.6040
1,000	11.6579	5.5577	6.1002
2,000	11.6581	5.5132	6.1449
3,000	11.6578	5.6036	6.0542
4,000	11.6578	5.5580	6.0998
Accurate Values	11.6573	5.6605	5.9968

good accuracy is not assured. Thus when $n = 800$, the naive binomial value is 5.052 when the true value is 5.661.

The situation is similar for the down-and-out call options, but in this case the naive binomial values lie above the true value. Once again, large n does not lead necessarily to an option value close to the true value.

More detailed calculations highlight an important pattern in the values of the two barrier options. This pattern is illustrated graphically in Exhibit 2, which plots the values of the barrier options for different numbers of time steps. The graphs show that the numerical values of the down-and-out call oscillate

late above the accurate values in sharply peaked waves of ever increasing length. At the trough of each wave the computed value is close to the accurate value.

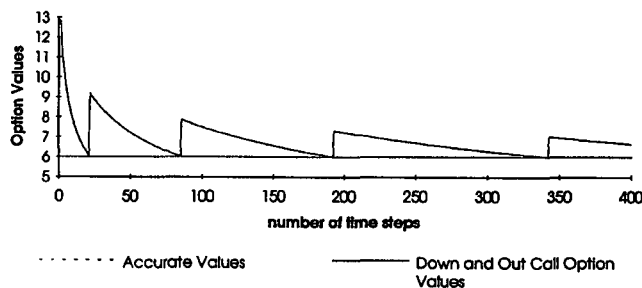
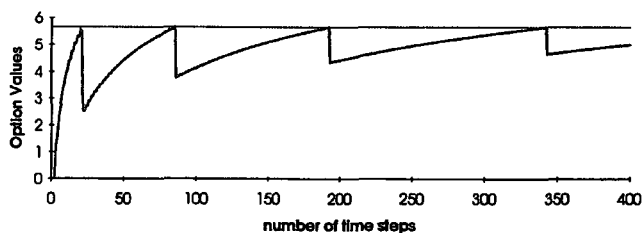
Similarly, the numerical values of the down-and-in call option oscillate below the accurate values in waves that are the mirror image of those for the down-and-out call option. At the crest of each wave, the computed value is close to the accurate value.

If a researcher is unaware of this pattern and uses the values of n suggested by the convergence rate for the standard call, the bias will be significant. Because these waves are mirror images of each other, the estimates for the standard call are much more stable and much more accurate as can be seen in the first column in Exhibit 1.² Good accuracy is obtained after 100 or so time steps, as the biases in the two barrier option values cancel each other out.

This pattern provides the clue to the origin of the poor convergence for the barrier options. In the binomial lattice, the barrier H will in general lie between two horizontal sets of nodes. This induces an inaccuracy into the calculations. As we increase the number of time steps n , the position of the horizontal

EXHIBIT 2

DOWN-AND-IN CALL OPTIONS AND DOWN-AND-OUT CALL OPTIONS FOR VARYING NUMBERS OF TIME STEPS



layers of constant asset prices in the tree changes, so that eventually there is a layer just beyond the barrier. When this happens we obtain a more accurate estimate of the barrier option.

We can arrange matters so that the n selected is optimal from the viewpoint of convergence. Using the notation of Cox, Ross, and Rubinstein, the amplitude of a downward jump d is given by

$$d = \exp(-\sigma\sqrt{h})$$

where $h = T/n$.

In general, the barrier H will lie between the asset price after m down jumps and the asset price after $(m + 1)$ down jumps so that

$$Sd^m > H > Sd^{m+1}$$

To arrange things so that the barrier is close to but just above a layer of horizontal nodes in the tree, we must select n so that it is the largest integer that is smaller than

$$F(m) = \frac{m^2 \sigma^2 T}{\left(\log \frac{S}{H}\right)^2} \quad m = 1, 2, 3, \dots$$

Note that because this n makes the size of the price move just large enough that m of them will take the price just beyond the barrier, the formula is the same for an "up" option, where $H > S$.

For different values of m , this formula generates a sequence of values of n that produce accurate values for the barrier options we have been considering. The first five values of $F(m)$ for the parameters used in Exhibit 1 are:

m	$F(m)$
1	21.38
2	85.52
3	192.42
4	342.08
5	534.51

In Exhibit 3 we use the values of n generated

EXHIBIT 3 NUMERICAL ESTIMATES OF STANDARD EUROPEAN CALL OPTION, DOWN-AND-IN CALL, AND DOWN-AND-OUT CALL BASED ON REVISED BINOMIAL METHOD

Barrier Level = 90% of Strike Price

Parameters

S	=	95
K	=	100
σ	=	0.25 per year
T	=	1 year
r	=	0.10 per year
n	=	number of time divisions in binomial method

n	Standard Call Option	Down-and-In Call Option	Down-and-Out Call Option
21	11.549	5.508	6.042
85	11.677	5.658	6.020
192	11.665	5.660	6.006
342	11.654	5.656	5.998
534	11.660	5.660	6.000
Accurate Values	11.6573	5.6605	5.9968

by this method to derive estimates for the two barrier options we have been considering. The estimates for the barrier options are now much more accurate, and errors are of the same order of magnitude as those for the standard call option.

II. CAPPED OPTIONS

Capped options are of both practical and theoretical interest. They have occurred as embedded options in corporate securities, and in 1991 the Chicago Board Options Exchange introduced capped index options on the S&P 100 and S&P 500. Flesaker [1992] pointed out some serious flaws in the design of the S&P caps and correctly predicted their ultimate demise.

Capped options are widespread in the interest rate market. Boyle and Turnbull [1989] explore some properties of European and American caps. Broadie and Detemple [1993] have recently published new closed-form solutions for different types of capped

options and analyzed certain features of these contracts.

There are a couple of reasons for revisiting the problem of pricing capped options in a binomial setting. First, many actual contracts are based on exercise at discrete time points, so that even if the continuous-time formula is available it may not be appropriate to use it, a point made also by Flesaker.

Second, we have found that some of the published numbers in the literature are somewhat biased. For example, the estimates for the American caps in the Boyle-Turnbull paper are produced using a finite difference approach that has the same type of bias we have seen in the case of down-and-out options.

Third, Broadie and Detemple point out the slow convergence rate of the binomial method in the discussion of their numerical results. Our discussion clarifies the source of the sluggish convergence and shows how to overcome it.

A European capped call option resembles a standard European call option, except that there is an upper limit on the payout. For example, if the limit is H , and $H > K$ (the strike price), then the option value at expiration is

Terminal State	Payoff
$S(T) > H$	$H - K$
$H > S(T) > K$	$S(T) - K$
$S(T) < K$	Zero

This contract can be viewed as the combination of a long European call with strike price K and a short European call with strike price H ; in other words, a vertical spread.

If the capped call is American, it may be optimal to exercise it early.³ In fact, Broadie and Detemple observe that the optimal strategy is to exercise the option as soon as the cap is reached.⁴ They exploit this insight to derive an analytical expression for the value of the American call. The American capped call can be expressed as an up-and-out European call with a rebate equal to the difference between the upper limit and the strike price.⁵

The formula can be rewritten as:

$$\begin{aligned} \text{CAC}[S, H, K, r, \sigma, T] = \\ \text{SN}(f(S, K)) - \text{Ke}^{-rT} \text{N}(f(S, K) - \sigma\sqrt{T}) \end{aligned}$$

$$\begin{aligned} & - \left\{ \text{SN}(f(S, H)) - \text{Ke}^{-rT} \text{N}(f(S, H) - \sigma\sqrt{T}) \right\} \\ & + \left\{ \frac{H}{S} \right\}^{\gamma_1} \left\{ \frac{H^2}{S} \text{N}(-f(H^2, SK)) - \text{Ke}^{-rT} \text{N}(-f(H^2, SK) \right. \\ & \left. - \sigma\sqrt{T}) \right\} - \left\{ \frac{H}{S} \right\}^{\gamma_1} \left\{ \frac{H^2}{S} \text{N}(-f(H, S)) \right. \\ & \left. - \text{Ke}^{-rT} \text{N}(-f(H, S) - \sigma\sqrt{T}) \right\} + (H - K) \\ & \times \left\{ \left\{ \frac{H^\gamma}{S^\gamma} \right\} \text{N}(-f(H, S)) + \left\{ \frac{S}{H} \right\} \text{N}(f(S, H)) \right\} \end{aligned}$$

where

$$\gamma = 2 \frac{r}{\sigma^2}$$

$$\gamma_1 = \frac{2r}{\sigma^2} - 1$$

$$f(A, B) = \frac{\left\{ \text{Log}\left(\frac{A}{B}\right) + \left(r + \frac{\sigma^2}{2}\right)T \right\}}{\sigma\sqrt{T}}$$

We can use this analytical expression to obtain exact values. The same modification discussed earlier enables us to obtain accurate values using the modified binomial method. The calculations are given in Exhibit 4. (The example is inspired by a capped oil indexed note.)

The modified binomial method produces capped option values that are quite accurate. The Boyle-Turnbull values underestimate the true values. If we use a standard binomial approach by selecting values of n in the conventional way ($n = 50, 100$, etc.), the resulting estimates would also be biased.

In Exhibit 4 we produce two sets of estimates, depending on the size of n , the number of the time steps. In both cases the size of n has been optimally selected. In the first set, the number of steps is around

EXHIBIT 4

VALUES OF AMERICAN CAPPED OPTIONS USING DIFFERENT VALUATION PROCEDURES

Parameters

S	=	13.5 to 40
K	=	25
H	=	40
σ	=	0.4 per year
T	=	4 years
r	=	$\log(1.08)$

S	Accurate Value (Broadie and Detemple)	Modified Binomial Method (Number of Time Steps)		Finite Difference Estimates (Boyle- Turnbull)
13.5	2.3534	2.3462 (65)	2.3512 (312)	2.2699
15	2.9569	2.9537 (66)	2.9513 (321)	2.8502
20	5.2025	5.2091 (65)	5.2022 (341)	5.0100
25	7.6351	7.6150 (72)	7.6300 (350)	7.3517
30	10.1183	10.1125 (69)	10.1147 (378)	9.7447
35	12.5837	12.5634 (35)	12.5866 (323)	12.1230
40	15.0000	15.0000	15.0000	15.0000

65, while in the second set it is around 350. Note that the accuracy is quite good for the smaller values of n . Step sizes of around 65 are attainable using a simple spreadsheet program.

It may be helpful to indicate how these optimal values of n are obtained. Take, for example, the case when $S = 25$. In this case the optimal values of n are generated by taking different values of m in the expression:

$$F(m) = \frac{m^2(0.4)^2 4}{\left\{\log \frac{25}{40}\right\}^2} \quad m = 1, 2, 3, \dots$$

Thus when $m = 5$, we obtain $F(5) = 72.43$, and when $m = 11$ we obtain $F(11) = 350.56$. This is where we get the number of time steps of 72 and 350 for $S = 25$ in Exhibit 4. For the other stock prices, the optimal values of n will be different, but we selected values close to 65 and 350 for two sets of results.

III. VULNERABLE OPTIONS

The standard option model assumes there is no default risk. Recently a number of authors have analyzed the impact of credit risk on the valuation of contingent claims. Johnson and Stulz [1987] derive valuation formulas for European options when there is a positive probability of default at expiration. Hull and White [1994] extend this model to allow for different types of recovery. Rich [1993b] develops a closed-form expression for certain types of vulnerable options.

Accurate numerical values can be obtained for vulnerable options if we incorporate the modification to the binomial method described earlier. If the naive binomial method is used, the values obtained are biased. A vulnerable option depends on the behavior of two risky assets, so we use the lattice procedure developed by Boyle [1988] to obtain numerical values.

We consider first a standard European call option on a foreign currency. The value of this option is:

$$C = Se^{-rT}N(d_1) - Ke^{-rT}N(d_1 - \sigma\sqrt{T})$$

where

$$d_1 = \frac{\left\{\log \frac{S}{K} + \left(r - r_f + \frac{\sigma^2}{2}\right)T\right\}}{\sigma\sqrt{T}}$$

and

- T = time to maturity of option;
- S = spot price of foreign currency;
- K = strike price of option;
- r = domestic riskless rate continuously compounded;
- r_f = foreign riskless rate continuously compounded; and
- σ = standard deviation of the exchange rate.

To introduce default risk, assume that the writer defaults if at any time during the life of the option the value of the writer's assets reaches a level H . Assume that the writer's assets follow a geometric

Brownian motion with standard deviation σ_A , and that the optioned asset S and the writer's assets A have a correlation coefficient ρ .

Rich obtains an expression for the value of this type of option:

$$C_V = e^{-r_f T} SM(y_1, y_2, \rho) - e^{-r_f T} K \times$$

$$M(y_1 - \sigma\sqrt{T}, y_2 - \sigma\rho\sqrt{T}, \rho) - \frac{H^\gamma}{A^\gamma} \left\{ e^{-r_f T} S \frac{H^\gamma}{A^\gamma} \times \right.$$

$$\left. M(y_3, y_4, \rho) + e^{-r_f T} KM(y_3 - \sigma\sqrt{T}, y_4 - \sigma\rho\sqrt{T}, \rho) \right\}$$

where

$$\gamma = 2 \frac{r}{\sigma_A^2} - 1$$

$$y_1 = \frac{\left\{ \log \frac{S}{K} + \left(r - r_f + \frac{\sigma^2}{2} \right) T \right\}}{\sigma\sqrt{T}}$$

$$y_2 = \frac{\left\{ \log \frac{A}{H} + \left(r - \frac{\sigma_A^2 - 2\sigma\sigma_A\rho}{2} T \right) \right\}}{\sigma_A\sqrt{T}}$$

$$y_3 = y_1 + \frac{2\rho \log \frac{H}{A}}{\sigma_A\sqrt{T}}$$

$$y_4 = y_2 + \frac{2 \log \frac{H}{A}}{\sigma_A\sqrt{T}}$$

and $M(a, b, \rho)$ is the standard bivariate cumulative normal density function with upper limits a and b and correlation coefficient ρ .

We can use this formula to obtain exact values of these vulnerable options. The Boyle [1988] lattice method can be used to obtain numerical estimates of such options. If we adopt a naive approach, we find that very large numbers of time steps are required to

obtain a high degree of accuracy. Indeed Rich finds that the numerical values derived by Hull and White are close to but consistently different from the accurate values. We can modify the Boyle lattice approach so that accurate results are obtained.

The Boyle lattice approach extends the Cox-Ross-Rubinstein method to handle payoffs that depend on two underlying risky assets. It is assumed that these two assets have a bivariate lognormal density. A discrete-time bivariate lattice is constructed so that the option values produced converge to their accurate continuous-time limits. The key idea is to construct a bivariate discrete-time probability distribution that has the desired means and variance-covariance parameters as the number of time steps is increased.

Suppose the current values of the two risky assets are S and A. At the end of a time step of length h ($= T/n$), there are five distinct outcomes. The five-point probability distribution of the asset prices one period later is

Probability	Asset 1	Asset 2
p_1	$S \exp(\sigma\sqrt{h})$	$A \exp(\lambda\sigma_A\sqrt{h})$
p_2	$S \exp(\sigma\sqrt{h})$	$A \exp(-\lambda\sigma_A\sqrt{h})$
p_3	$S \exp(-\sigma\sqrt{h})$	$A \exp(\lambda\sigma_A\sqrt{h})$
p_4	$S \exp(-\sigma\sqrt{h})$	$A \exp(-\lambda\sigma_A\sqrt{h})$
p_5	S	A

Explicit expressions for the probabilities are given in Boyle [1988]. The parameter λ is chosen by the user. We use a value of 1.2 based on the earlier findings of Boyle.

This five-point modular probability distribution can be used to construct a three-dimensional binomial lattice. Each node is identified by its time coordinate and two price coordinates, one corresponding to each risky asset. To value a European contract we can specify the value of the security at maturity and obtain its current value by backward recursion through the lattice in the usual way.

In the application to vulnerable options, we have to recognize that the option value is zero if ever the state variable A hits the barrier H. If we do not preselect the number of time steps, the discretization

EXHIBIT 5

COMPARISON OF VULNERABLE EUROPEAN CALL OPTION VALUES OBTAINED BY ACCURATE FORMULA AND MODIFIED LATTICE APPROACH

Parameters

S	=	100
A	=	100
K	=	100
H	=	94 and 96
σ	=	0.15 per year
σ_A	=	0.05 per year
r	=	-0.8 to 0.8
T	=	1 year
r	=	0.05 per year
r_f	=	0.05 per year

Small "n" Values Used:

n = 60	for H = 94:	n = 54	for H = 96
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Large "n" Values Used:

n = 304	for H = 94:	n = 311	for H = 96
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r	H	Accurate Value (Rich Formula)	Modified Lattice Values	
			n = 60	n = 304
-0.8	94	4.577	4.574	4.577
-0.4	94	5.069	5.067	5.069
0.0	94	5.409	5.408	5.409
0.4	94	5.605	5.604	5.605
0.8	94	5.677	5.675	5.677
			n = 54	n = 311
-0.8	96	3.287	3.275	3.285
-0.4	96	4.164	4.156	4.163
0.0	96	4.827	4.825	4.827
0.4	96	5.298	5.304	5.299
0.8	96	5.569	5.582	5.571

procedure for A implies that H will lie between adjacent layers of A values. The same type of bias we described earlier will reappear.

Our earlier discussion leads us to the remedy. In this case we need values of n so that the A values are below but as close as possible to H. The appropriate number of time steps is given by the integral values of

$$F(m) = \frac{m^2 \lambda^2 \sigma_A^2 T}{\left\{ \log \frac{A}{H} \right\}^2} \quad m = 1, 2, 3 \dots$$

We use two sets of values of n in Exhibit 5 to compute the values of vulnerable options. The accurate values are obtained from Rich's formula. Note that the modified lattice method produces excellent results even for fifty or sixty time steps if n is chosen appropriately.

We also used larger values of n (around 300). With these values the modified lattice values are very close to the accurate values, and in many cases indistinguishable from them to three decimal places.

IV. CONCLUSION

The standard binomial method converges very slowly when it is used to value barrier options, but we show how to modify the standard method to overcome this drawback. Our modification is simple to apply. We demonstrate its effectiveness by valuing three types of contracts for which analytical solutions exist: simple barrier options, capped American options, and vulnerable options.

Because our modified approach gives a very high degree of accuracy, it should have practical value, given the importance of barriers and caps in option contracts.

ENDNOTES

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¹The level of accuracy depends principally on the volatility of the basic asset and the time to option maturity. We see from Exhibit 1 that the Black-Scholes value for a one year at-the-money call with a 25% volatility level is \$11.657 when the current asset is \$100. The binomial method produces a value of \$11.634 with a hundred time steps. The difference is about 0.25% of the true option value. We consider this level of agreement accurate, although more precision may sometimes be desirable.

²Recall that the standard call is a combination of the down-and-in call and the down-and-out call.

³This is in marked contrast to the standard (uncapped) call, where in the absence of dividends it is never optimal to exercise an American option prior to maturity.

⁴Here we assume no dividends. Broadie and Detemple analyze the case where the underlying asset

pays dividends.

⁵The term, "rebate," in the barrier option literature denotes a terminal payment made under the option contract when the underlying asset value hits the barrier before expiration. In the examples discussed in Section I, the rebate was zero.

REFERENCES

- Boyle, Phelim P. "A Lattice Framework for Option Pricing with Two State Variables." *Journal of Financial and Quantitative Analysis*, 35, 1 (1988), pp. 1-12.
- Boyle, Phelim P., and Stuart M. Turnbull. "Pricing and Hedging of Capped Options." *Journal of Futures Markets*, 9, 1 (1989), pp. 41-54.
- Broadie, Mark, and Jerome Detemple. "American Capped Call Options on Dividend Paying Assets." Working Paper, Graduate School of Business, Columbia University, New York, 1993.
- Cox, John C., and Mark Rubinstein. *Options Markets*. Englewood Cliffs, NJ: Prentice-Hall, Inc., 1985.
- Flesaker, Bjorn. "The Design and Valuation of Capped Stock Index Options." Working Paper, University of Illinois at Urbana-Champaign, 1992.
- Hull, John, and Alan White. "The Impact of Default Risk on the Prices of Options and Other Derivative Securities." *Journal of Banking and Finance*, forthcoming, 1994.
- Johnson, Herb, and René Stulz. "The Pricing of Options with Default Risk." *Journal of Finance*, 62 (1987), pp. 267-280.
- Merton, Robert C. "Theory of Rational Option Pricing." *Bell Journal of Economics and Management Science*, 4 (1973), pp. 141-183.
- Rich, Don. "The Mathematical Foundations of Barrier Option Pricing Theory." Working Paper, Virginia Polytechnic Institute, 1993a.
- . "The Valuation of Black Scholes Options Subject to Intertemporal Default Risk." Working Paper, Virginia Polytechnic Institute, 1993b.
- Rubinstein, Mark, and Eric Reiner. "Exotic Options." Working Paper, University of California at Berkeley, 1993.